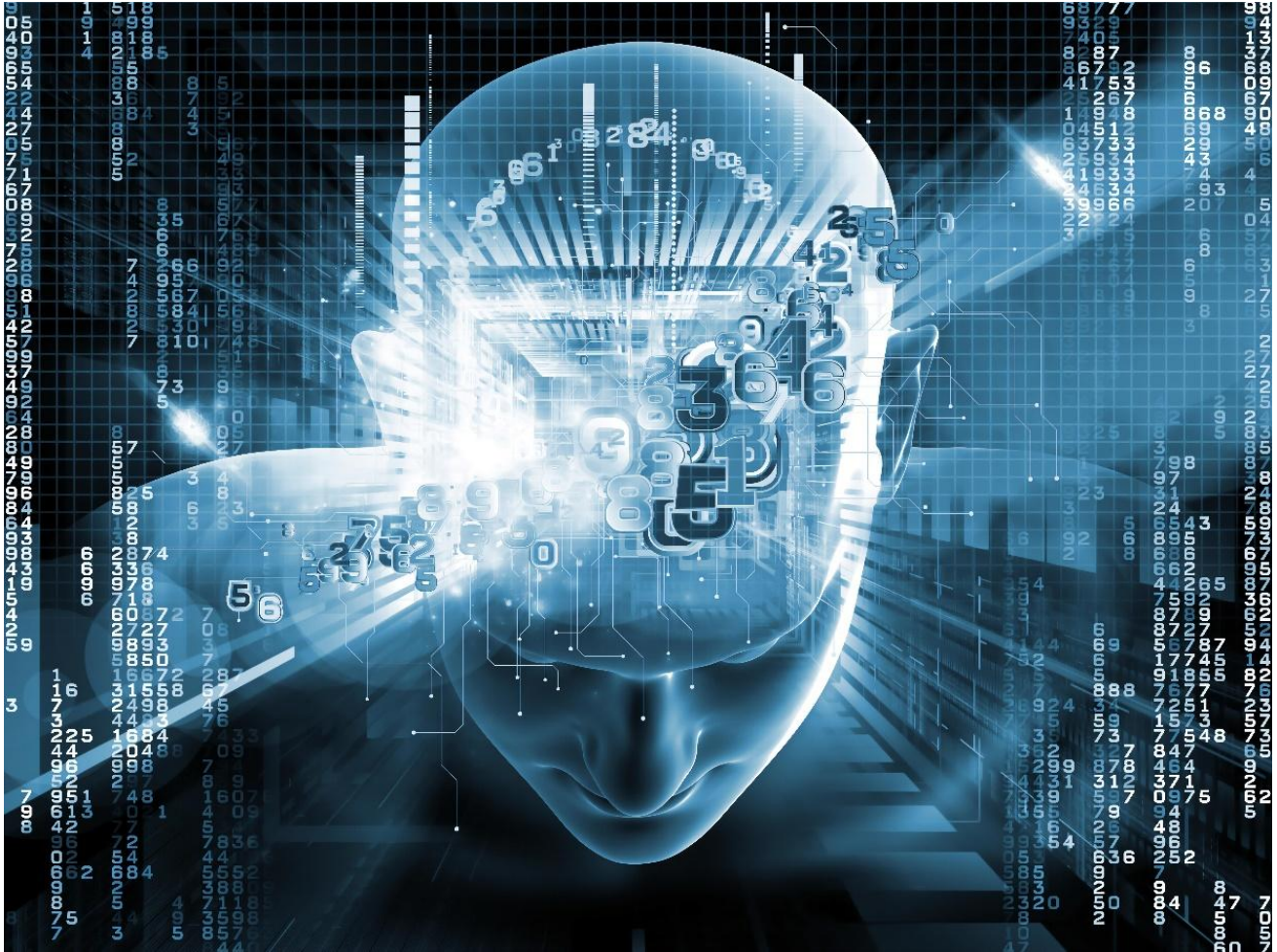




Claude for Finance – Excel Session 2





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Training Course Documentation

Excel AI Skills — Session 2 Practical Exercises

Overview

This document covers the complete build process for transforming the starting workbook (6 sheets) into the solution workbook (17 sheets) across 4 phases.

Starting File: 6 sheets — Workbook Overview, Read Me, Fact_Financials (OG/New/Rolled Act), Dynamic Extracts

Solution File: 17 sheets — Starting + 11 new sheets built using Claude AI skills

Data Structure: tbl_Financials table with 7,812 rows × 6 columns (DateKey, EntityKey, AccountKey, DeptKey, ScenarioKey, Amount). 3 entities (Americas, APAC, EMEA), 8 accounts, 3 scenarios (Actuals/Budget/Forecast), 36 months (Jan 2024 – Dec 2026).

Key Infrastructure: 3 named LAMBDA's on Dynamic Extracts:

- PLSTRUCTURE() — returns sorted P&L account labels (spills)
- PLDATA(dates, scenario) — consolidated amounts per account per date (spills)
- PLDATABY(dates, scenario, entity_cell) — entity-filtered amounts (spills)

Phase 1: Data Validation & Reconciliation

Skill: /training-data-reconciliation /Sheet Built: Validation Checks

What Was Built

A 57-check validation sheet with formula-driven PASS/FAIL checks across 8 sections:

1. Data Integrity (10 checks): Row counts, column counts, blank detection, unique dimension counts, date coverage, OG-vs-New comparison
2. Roll-Forward Cross-Check (6 checks): Scenario key preservation, actuals gained, budget unchanged, forecast reduced, total rows preserved, zero-sum roll
3. Table Conversion (4 checks): tbl_Financials row/column count, no blanks in Amount, all ScenarioKeys present
4. Management Report Tie-Out (8 checks): Table ↔ Dynamic tie, consolidation checks, source tie, period counts
5. Actual Period Integrity (6 checks): Non-zero month counts, last actual month, Aug–Dec zeroed
6. Source Data Integrity (8 checks): Account-level and entity-level SUMIFS vs report totals
7. Dynamic Report Cross-Checks (6 checks): Entity NP tie, variance integrity, entity count
8. Driver Forecast Integrity (5 checks): Historical tie, P&L arithmetic, consolidation, output tie, toggle response

Key Formula Patterns

- Row count: =ROWS('Fact_Financials- OG'!B4:B7815)
- Unique count: =ROUND(SUMPRODUCT(1/COUNTIF(range,range)),0)
- Difference check: =SUMPRODUCT(('Sheet1'!range<>'Sheet2'!range)*1)
- Status: =IF(Expected=Actual,"PASS","FAIL")
- Section summary: =COUNTIF(status_range,"PASS")&" of "&COUNTA(#_range)&" checks passed"
- Conditional formatting: PASS = green (#C4DDD2 fill, #006100 font), FAIL = red (#FFC7CE fill, #9C0006 font)



Design Decisions

- Expected values are hardcoded (blue font) as reference benchmarks
- Actual values are always formulas — never hardcode the check result
- Each section has a summary row counting PASS/FAIL
- Overall summary at top: "X of 57 checks passed"

Phase 2: Management Report Creation

Skill: /training-management-report-build | Sheets Built: Mgmt Report — Table, Mgmt Report — Dynamic

What Was Built

Two P&L management reports producing identical output with different formula approaches:

Table Report (SUMIFS approach)

- 4 entity blocks × 3 views (Actuals/Budget/Variance) = 12 sections
- 12 P&L line items per section: Product Revenue, Services Revenue, Total Revenue, COGS, Gross Profit, S&W, Marketing, Software, T&E, Other OpEx, Total OpEx, Net Profit
- 36 monthly columns (Jan 2024 – Dec 2026)
- Core formula: =SUMIFS(tbl_Financials[Amount], EntityKey filter, AccountKey filter, ScenarioKey filter, DateKey filter)
- Consolidation uses EntityKey="*" wildcard

Dynamic Report (LAMBDA approach)

- Same output structure but using spilling LAMBDA's
- PLSTRUCTURE() for row labels (auto-updates if accounts are added)
- PLDATA(dates,scenario) for consolidated data (one formula → entire block)
- PLDATABY(dates,scenario,entity) for entity-filtered data
- Variance via array subtraction: =Actuals# - Budget#

Teaching Points

- SUMIFS is explicit but verbose — one formula per cell, manually managed
- LAMBDA's are concise but abstract — one formula spills an entire block
- Both produce identical numbers — the difference is maintainability and scalability
- LAMBDA's auto-expand when new accounts or periods are added
- SUMIFS requires manual range extension for new data

Key Formula Examples

Table:

=SUMIFS(tbl_Financials[Amount],tbl_Financials[EntityKey],"*",tbl_Financials[AccountKey],\$B7,tbl_Financials[ScenarioKey],1,tbl_Financials[DateKey],C\$3)

Dynamic: =PLDATA(C\$6#,1) — one formula replaces 432 SUMIFS formulas (12 rows × 36 cols)



Phase 3: Forecast Accuracy Reporting

Skill: /training-forecast-accuracy-report / **Sheet Built:** Forecast Accuracy Report

What Was Built

H1 CY2026 (Jan–Jun 2026) comparison of Actuals vs Budget vs Forecast across all 4 entity groups.

Key Innovation: Uses FILTER to isolate the 6-month overlap period:

```
=FILTER(TRANSPOSE(SORT(UNIQUE(tbl_Financials[DateKey]))), dates>=20260131)
```

Each entity block contains: monthly Actuals, Budget, Forecast data + H1 Total + \$ Variance + Accuracy %.

Accuracy Formula

```
=IFERROR(1-ABS(Variance/Budget),0)
```

- 100% = perfect forecast
- 95%+ = green (excellent)
- <80% = red (needs investigation)

Teaching Points

- Forecast accuracy is measured on the OVERLAP period only (where actuals exist)
- FILTER + PLDATA combination isolates dates dynamically
- IFERROR prevents #DIV/0! on zero-budget lines
- Accuracy is always positive (ABS) — direction comes from the variance sign



Phase 4: Scenario Analysis & Executive Summary

Skill: /training-scenario-executive-summary/Sheets Built: Scenario Manager, Driver Analysis, Forecast Output, Scenario Dashboard, Executive Summary

What Was Built

Scenario Manager

Input sheet with 11 growth drivers across 3 scenarios (Base/Upside/Downside):

- Revenue: Volume growth, price escalation (Product + Services)
- Costs: COGS %, S&W growth, marketing %, software/T&E/other growth
- Active column responds to Dashboard toggle via INDEX/MATCH
- Seed values: Jul 2026 actual volumes, prices, FTE by entity

Driver Analysis

Volume × Rate decomposition engine:

- 18-month window: Feb 2026 – Jul 2027 (6 actuals + 12 forecast)
- 3 entity blocks + consolidated
- Revenue = Volume × Unit Price; Services = Hours × Rate
- Costs: COGS as % of revenue, S&W as FTE × salary with growth, etc.
- Compound monthly growth: $\text{prior_month} \times (1 + \text{annual_rate}/12)$

Forecast Output

- Monthly P&L linked to Driver Analysis
- Consolidated figures from Driver Analysis
- Scenario comparison: Budget vs Original Forecast vs New Forecast

Scenario Dashboard

- Toggle in T22 (Base/Upside/Downside) controls which scenario is active
- Revenue + Net Income comparison charts (4 series each)
- Y-axes locked for fair cross-scenario comparison

Executive Summary

- KPIs: FY26 Revenue, Gross Margin %, Net Profit, Budget/Forecast Accuracy
- FY P&L: FY25 vs FY26 Actuals vs Budget with YoY and variance
- Entity breakdown: Revenue + NP by Americas, APAC, EMEA
- 4 charts: Revenue mix, P&L waterfall, entity comparison, budget trend



Teaching Points

- Assumptions MUST be separated from formulas (Scenario Manager holds all inputs)
- Driver decomposition reveals WHAT drives the P&L, not just what it totals to
- Scenario toggle enables instant what-if analysis without rebuilding
- Executive Summary aggregates everything into one board-ready view
- All forecast cells reference Scenario Manager — change one assumption, entire model updates

Summary of Skills Created

	Skill	Sheets	Key Formulas
1	/training-data-reconciliation	Validation Checks	SUMIFS, COUNTIFS, ROWS, SUMPRODUCT
2	/training-management-report-build	Mgmt Report Table + Dynamic	SUMIFS vs PLDATA/PLDATABY
3	/training-forecast-accuracy-report	Forecast Accuracy Report	FILTER + PLDATA + accuracy %
4	/training-scenario-executive-summary	5 sheets (Scenario → Exec Summary)	INDEX/MATCH, compound growth, SUMIFS

Workbook Architecture

Starting: 6 sheets → Solution: 17 sheets

- 57 validation checks (all PASS)
- 4 entity groups × 3 scenarios
- 36-month historical + 12-month forecast
- 100% formula-driven (zero hardcoded outputs)



Complete Prompt Playbook — Starting File to Solution File

This section contains every prompt a trainee needs to transform the starting workbook (6 sheets) into the completed solution (17 sheets). Prompts 1–2 (Data Import Methods) are already on the Read Me sheet. Prompts 3–13 below cover the full build journey.

Each prompt is self-contained — it includes all the context Claude needs (sheet names, cell references, column layouts, expected outputs) so there is no ambiguity. Copy-paste the full prompt text (shown in the indented blocks below) into Claude for each step.

Prompt 3 — Data Validation & Reconciliation (57 Checks)

Important context about this workbook:

tbl_Financials is the working data table on 'Fact_Financials- New' with columns: DateKey (YYYYMMDD integer), EntityKey (Americas/EMEA/APAC), AccountKey (8 accounts), DeptKey, ScenarioKey (1=Actual, 2=Budget, 3=Forecast), Amount. Coverage: 36 months (Jan 2024 – Dec 2026), 3 entities, 8 accounts, 3 scenarios, 7,812 rows.

The workbook also has 'Fact_Financials- OG' (original pre-roll data) and 'Fact_Financials- Rolled Act' (the transformation source). The 'Dynamic Extracts' sheet has 3 named LAMBDA: PLSTRUCTURE(), PLDATA(dates, scenario), PLDATABY(dates, scenario, entity). Two management reports already exist: 'Mgmt Report – Table' (SUMIFS) and 'Mgmt Report – Dynamic' (LAMBDA).

Task — Build 57-check Validation Sheet:

Build a new sheet called 'Validation Checks' with 57 automated PASS/FAIL checks across 8 sections. Each check has 4 columns: Check Description, Expected (blue font #0000FF hardcoded reference value), Actual (formula – NEVER hardcode), Status =IF(Expected=Actual,"PASS","FAIL"). Each section ends with a summary row: =COUNTIF(status_range,"PASS")&" of "&COUNTA(check_range)&" checks passed". An overall summary at the top: "X of 57 checks passed".

Section 1 – Data Integrity (10 checks): Row count on OG sheet =ROWS(range), row count on New sheet, column count (6), blank cells in Amount =COUNTBLANK(), unique EntityKeys =ROUND(SUMPRODUCT(1/COUNTIF(range,range)),0), unique AccountKeys, unique ScenarioKeys, unique DateKeys, first DateKey =MIN(), last DateKey =MAX(). Expected values: 7812 rows, 6 columns, 0 blanks, 3 entities, 8 accounts, 3 scenarios, 36 dates.

Section 2 – Roll-Forward Cross-Check (6 checks): ScenarioKey preservation (OG scenario counts = New scenario counts), Actuals gained =COUNTIFS(New,ScenarioKey=1) minus COUNTIFS(OG,ScenarioKey=1), Budget unchanged =SUMPRODUCT((OG_budget<>New_budget)*1) should be 0, Forecast reduced =COUNTIFS(New,3) < COUNTIFS(OG,3), total rows preserved (New = OG), zero-sum roll =SUM(New Amount) - SUM(OG Amount) difference check.

Section 3 – Table Conversion (4 checks): tbl_Financials row count =ROWS(tbl_Financials[Amount]), column count =COLUMNS(tbl_Financials), no blanks =COUNTBLANK(tbl_Financials[Amount]), all 3 ScenarioKeys present =ROWS(UNIQUE(tbl_Financials[ScenarioKey])).

Section 4 – Management Report Tie-Out (8 checks): Table Group Actuals NP total = Dynamic Group Actuals NP total, Table entity NP = Dynamic entity NP for each of Americas/APAC/EMEA, Group NP = SUM(entity NPs) on Table report, Group NP = SUM(entity NPs) on Dynamic report, source tie =SUMIFS(tbl_Financials[Amount],ScenarioKey,1) = report total, period count = 36.

Section 5 – Actual Period Integrity (6 checks): Count of months with non-zero Actuals, last actual month DateKey, Aug 2026 Actuals = 0, Sep 2026 = 0, Oct 2026 = 0, Nov-Dec 2026 = 0. Use SUMIFS filtered by DateKey and ScenarioKey=1.



Section 6 – Source Data Integrity (8 checks): Per-account SUMIFS vs report row totals for Product Revenue, Services Revenue, COGS, S&W (4 account-level checks). Per-entity source tie for Americas, APAC, EMEA, plus consolidated wildcard "*" (4 entity-level checks). All use SUMIFS against tbl_Financials.

Section 7 – Dynamic Report Cross-Checks (6 checks): Entity NP tie (Americas Dynamic NP = Table NP, APAC, EMEA), Variance integrity (Actuals minus Budget = Variance for each entity on Dynamic report), entity count = 3.

Section 8 – Driver Forecast Integrity (5 checks): Historical actuals tie (Driver Analysis actuals = tbl_Financials actuals for overlap period), P&L arithmetic (Revenue + COGS = GP, GP + OpEx = NP), consolidation (Consolidated = sum of 3 entities), Output tie (Forecast Output NP = Driver Analysis Consolidated NP), scenario toggle response (changing Scenario Dashboard toggle changes Active column in Scenario Manager).

Formatting: Section headers dark green (#1A3C2E) with white bold text. PASS cells green fill (#C4DDD2) with dark green font (#006100). FAIL cells red fill (#FFC7CE) with dark red font (#9C0006). Expected values blue font (#0000FF). Freeze top rows. Column widths: Description 350pt, Expected/Actual/Status 100pt each.

Prompt 4 — Management Report (Table Approach Using SUMIFS)

Important context about this workbook:

tbl_Financials is the working data table on 'Fact_Financials- New' with columns: DateKey (YYYYMMDD integer), EntityKey (Americas/EMEA/APAC), AccountKey (8 accounts), DeptKey, ScenarioKey (1=Actual, 2=Budget, 3=Forecast), Amount (positive=revenue, negative=expense). Coverage: 36 months (Jan 2024 – Dec 2026), 3 entities, 8 accounts, 3 scenarios. Net Profit = SUM(Amount) with no sign flip.

Task — Build Management Report using SUMIFS:

Build a new sheet called 'Mgmt Report – Table' with a P&L management report using SUMIFS structured references against tbl_Financials.

1. Date Row: Create a DateKey row with all 36 monthly DateKey values (20240131, 20240229, ..., 20261231). Below it, add a display row with formatted dates using TEXT(LEFT(datekey,4),MID(datekey,5,2),RIGHT(datekey,2)),"mmm-yy").
2. Build 4 entity blocks stacked vertically: Group Consolidated (EntityKey="*" wildcard), Americas, APAC, EMEA.
3. Each entity block has 3 sections: Actuals (ScenarioKey=1), Budget (ScenarioKey=2), Variance (= Actuals minus Budget for each cell).
4. Row labels for each section (12 rows): Product Revenue, Services Revenue, Total Revenue (=sum of above 2), COGS, Gross Profit (=Total Revenue + COGS since COGS is negative), Salaries & Wages, Marketing, Software & Tools, T&E, Other OpEx, Total Operating Expenses (=sum of 5 OpEx lines), Net Profit (=Gross Profit + Total OpEx).
5. Core SUMIFS formula pattern: =SUMIFS(tbl_Financials[Amount], tbl_Financials[EntityKey],\$entity, tbl_Financials[AccountKey],\$account_label, tbl_Financials[ScenarioKey],\$scenario_key, tbl_Financials[DateKey],date_key_ref). For Group Consolidated use EntityKey="*". Subtotal and total rows use SUM of component rows, NOT SUMIFS.



6. Formatting: Entity headers dark green (#1A3C2E) white bold text. Section headers medium green (#2D6A4F). Subtotals bold with light green (#D8F3DC) fill. Net Profit bold with double top border. Numbers #,##0;(##0);"-". Label column 200pt, data columns 85pt. Freeze header rows and label column.

7. Validation: Sum of all Actuals Net Profit must equal `SUMIFS(tbl_Financials[Amount],tbl_Financials[ScenarioKey],1)`. Group NP must equal Americas + EMEA APAC for each month.

Prompt 5 — Management Report (Dynamic Approach Using LAMBDA's)

Important context about this workbook:

`tbl_Financials` exists on 'Fact_Financials- New'. The Dynamic Extracts sheet has 3 named LAMBDA's already registered: `PLSTRUCTURE()` returns sorted unique account list (P&L row labels). `PLDATA(date_array, scenario_key)` returns consolidated P&L data for all accounts x dates. `PLDATABY(date_array, scenario_key, entity_cell)` returns entity-filtered P&L data. These produce spilling arrays.

Task — Build Dynamic Management Report using LAMBDA's:

Build a new sheet called 'Mgmt Report – Dynamic' that produces the SAME output as 'Mgmt Report – Table' using dynamic array formulas.

1. Date Array (Row 6): `C6 = TRANSPOSE(SORT(UNIQUE(tbl_Financials[DateKey])))` – spills 36 values.
2. Display Dates (Row 7): `C7 = MAP(C6#, LAMBDA(dk, TEXT(LEFT(dk,4),MID(dk,5,2),RIGHT(dk,2)),"mmm-yy")))`
3. Consolidated blocks: Row labels `B10 = PLSTRUCTURE()` (spills down). Actuals `C10 = PLDATA(C$6#,1)` (spills entire grid). Budget `C23 = PLDATA(C$6#,2)`. Variance `C36 = C10# - C23#` (array subtraction).
4. Entity blocks: `B49 = "Americas"` (entity ref). `C49 = PLDATABY(C$6#,1,$B$49)`. Budget `C62 = PLDATABY(C$6#,2,$B$49)`. Variance `C75 = C49# - C62#`. Repeat for EMEA and APAC.
5. CRITICAL: ONE formula per section fills the entire grid. Do NOT put anything in the spill range or you get #SPILL! errors. Output MUST match the Table report exactly.
6. Formatting: Same as Table report. Apply AFTER formulas spill.
7. Validation: SUM of Dynamic Group Actuals NP must equal SUM of Table Group Actuals NP (difference = 0).



Prompt 6 — Forecast Accuracy Report (H1 CY2026)

Important context:

tbl_Financials has Actuals (ScenarioKey=1), Budget (2), and Forecast (3). H1 CY2026 = January through June 2026 ONLY (6 months, DateKeys 20260131 to 20260630). Do NOT include July. LAMBDA's PLSTRUCTURE, PLDATA, PLDATABY are available.

Task — Build H1 CY2026 Forecast Accuracy Report:

Build a new sheet called 'Forecast Accuracy Report'.

1. Date Isolation: Filter to exactly 6 months (20260131, 20260228, 20260331, 20260430, 20260531, 20260630). IMPORTANT: H1 means January through June ONLY.
2. Build 4 entity blocks: Consolidated, Americas, APAC, EMEA.
3. Each block has 7 sections: (a) Monthly Actuals via PLDATA/PLDATABY for ScenarioKey=1, (b) Monthly Budget ScenarioKey=2, (c) Monthly Forecast ScenarioKey=3, (d) H1 Total = SUM of 6 months, (e) Budget Variance = H1 Actuals minus H1 Budget, (f) Budget Accuracy = $1 - \text{ABS}(\text{Variance} / \text{Budget})$, (g) Forecast Variance and Accuracy same pattern.
4. Guard against #DIV/0! with IFERROR on accuracy formulas.
5. Formatting: Accuracy as 0.0%. $\geq 95\%$ green highlight (#C4DDD2). $< 80\%$ red (#FFC7CE). Variance as #,##0;(#,##0);"-".
6. Validation: H1 Actuals Total Revenue should equal $\text{SUMIFS}(\text{tbl_Financials}[\text{Amount}], \text{ScenarioKey}, 1, \text{DateKey}, ">=20260131", \text{DateKey}, "<=20260630")$ filtered to revenue accounts only.

Prompt 7 — Scenario Manager

Important context:

tbl_Financials has 31 months of actuals (Jan 2024 – Jul 2026). The last actual month is July 2026. Forecast starts August 2026.

Task — Build Scenario Manager:

Build a new sheet called 'Scenario Manager' with 3 scenarios (Base/UpSide/Downside).

1. Revenue Drivers: Product Volume Growth (Base=4.5%, Up=6%, Down=2%), Price Escalation (2.5%/3%/1%), Service Volume Growth (4.5%/6%/2%), Service Rate Escalation (1.5%/2%/0.5%).
2. Cost Drivers: COGS % of Revenue (48%/45%/54%), S&W Growth (-2%/-3%/-0.5%), Marketing % of Revenue (8%/6%/12%), Software Growth (0.4%/0%/0.5%), T&E Growth (0.3%/0%/0.3%), Other OpEx Growth (0.3%/0%/0.3%).
3. Seed Values: July 2026 actuals by entity extracted from tbl_Financials using SUMIFS for DateKey=20260731, ScenarioKey=1. Include Product Units, Unit Price, Service Hours, Hourly Rate, FTE count, Salary/FTE for Americas, EMEA, APAC.
4. Active Column: F7 = $\text{INDEX}(C7:E7, \text{MATCH}(\text{'Scenario Dashboard'!T22, \{"Base", "Upside", "Downside"\}}, 0))$. Copy down for all drivers.
5. Formatting: Blue font (#0000FF) on all input cells. Yellow fill on assumptions. Active column light green (#D8F3DC).



Prompt 8 — Driver Analysis (Volume × Rate Decomposition)

Important context:

Scenario Manager has growth rates and seed values. July 2026 is the last actual month.

Task — Build Driver Analysis:

Build 'Driver Analysis' with 18-month Volume × Rate decomposition (Feb 2026 – Jul 2027). 6 months actuals + 12 months forecast.

1. Structure: 4 entity blocks – Americas, EMEA, APAC, Consolidated (= sum of 3).
2. Revenue: Product Revenue = Volume × Unit Price, Services Revenue = Hours × Rate. Actuals seed from tbl_Financials SUMIFS. Forecast = prior month × (1 + Scenario Manager active growth / 12).
3. Costs: COGS = Total Revenue × COGS% × -1. S&W = FTE × Salary/FTE with monthly growth. Marketing = Revenue × Mkt% × -1. Software/T&E/Other = prior × (1 + growth/12).
4. P&L: Total Revenue, Gross Profit (= Rev + COGS), Total OpEx, Net Profit (= GP + OpEx).
5. Consolidated = Americas + EMEA + APAC for each line.
6. CRITICAL: Divide annual growth rates by 12 for monthly compounding. COGS and expenses are NEGATIVE.

Prompt 9 — Forecast Output and Scenario Dashboard

Task — Build Forecast Output and Scenario Dashboard:

PART A – Forecast Output: Build 'Forecast Output' linking to Driver Analysis consolidated. Full P&L with same dates. Add Scenario Comparison section: Budget (SUMIFS ScenarioKey=2) vs Original Forecast (ScenarioKey=3) vs New Forecast (from Driver Analysis).

PART B – Scenario Dashboard: Build 'Scenario Dashboard' with scenario toggle dropdown in T22 (Base/UpSide/Downside). Two charts: Revenue Comparison and Net Income Comparison, each with 4 series (Actuals, Budget, Original Forecast, New Forecast). Lock Y-axes across scenarios.



Prompt 10 — Executive Summary

Task — Build Executive Summary:

Build 'Executive Summary' as a board-ready dashboard.

1. KPI tiles: FY26 Revenue (SUMIFS DateKey>=20250701 AND <=20260630), Gross Margin %, Net Profit, Budget Accuracy % (from Forecast Accuracy Report), Forecast Accuracy % (from Forecast Accuracy Report).
2. FY P&L Summary: FY25 vs FY26 Actuals vs FY26 Budget with YoY and Budget variance.
3. Entity Breakdown: Revenue and NP by entity with % of total.
4. Charts: Revenue Mix pie, P&L Waterfall, Entity Comparison bar, Actual vs Budget Trend line.
5. Commentary section with formula-driven IF insights.

Prompt 11 — Validation Checks (57 Checks Across 8 Sections)

Task — Build comprehensive Validation Checks:

Build 'Validation Checks' with 57 automated PASS/FAIL checks.

8 sections: (1) Data Integrity – row counts, blank checks, dimension counts. (2) Roll-Forward Cross-Check – OG vs New vs Rolled Act. (3) Table Conversion – tbl_Financials structure. (4) Management Report Tie-Out – Table equals Dynamic, consolidation, source tie. (5) Actual Period Integrity – non-zero month counts, last actual month, future zeroes. (6) Source Data Integrity – account-level and entity-level SUMIFS vs reports. (7) Dynamic Report Cross-Checks – entity NP tie, variance integrity. (8) Driver Forecast Integrity – historical tie, P&L arithmetic, consolidation.

Formula patterns: Expected (blue hardcoded), Actual (ROWS/SUMIFS/COUNTIF formula), Status = IF(Expected=Actual,"PASS","FAIL"). Conditional formatting: PASS green (#C4DDD2), FAIL red (#FFC7CE).

Prompt 12 — Create Reusable Skills

Task — Create 4 Claude skills:

1. /training-data-reconciliation – Builds the 57-check Validation Checks sheet. Include trigger phrases, prerequisites, formula patterns, formatting specs, and gotchas.
2. /training-management-report-build – Builds both Table (SUMIFS) and Dynamic (LAMBDA) P&L reports. Capture the exact formula patterns for both approaches.
3. /training-forecast-accuracy-report – Builds H1 CY2026 accuracy comparison with PLDATA/PLDATBY and FILTER. Emphasise the 6-month H1 boundary.
4. /training-scenario-executive-summary – Builds the 5-sheet scenario/forecast/dashboard/executive package. Capture the interconnections between sheets.



Prompt 13 — Cross-Validation Against Solution

Task — Cross-validate your build:

Build a 'Cross-Validation' sheet with 34 checks across 6 sections comparing your workbook against the solution file. Sections: Source Data Integrity (5 checks), Table Report (8), Dynamic Report (4), Forecast Accuracy (6), Scenario Manager (6), Executive Summary (5). Each check: Solution Value, Your Value, Variance, Status. This proves your build matches the reference and highlights any interpretation differences.

Session Flow Summary

Step	Prompt	Sheet(s) Created	Skill Created
1	Prompts 1–2 (on Read Me)	tbl_Financials, Dynamic Extracts	—
2	Prompt 3	Validation Checks	—
3	Prompt 4	Mgmt Report — Table	—
4	Prompt 5	Mgmt Report — Dynamic	—
5	Prompt 6	Forecast Accuracy Report	—
6	Prompt 7	Scenario Manager	—
7	Prompt 8	Driver Analysis	—
8	Prompt 9	Forecast Output, Scenario Dashboard	—
9	Prompt 10	Executive Summary	—
10	Prompt 11	Validation Checks	—
11	Prompt 12	—	4 skills
12	Prompt 13	Cross-Validation	—

Total: 13 prompts → 17 sheets + 4 skills = complete solution



Short Form Prompts

Short form prompts (if you don't have or have not built out the skills depth we have) will work eventually but you have to have the foundations to build it on.

Here are the equivalent prompts used in the training now.

Prompt S1 — Validation Checks

Build a 'Validation Checks' sheet with 53 automated PASS/FAIL checks across 8 sections: (1) Data Integrity — 10 checks on row counts, columns, blanks, unique dimensions, date range across the Fact_Financials sheets. (2) Roll-Forward Cross-Check — 6 checks comparing OG vs New. (3) Table Conversion — 4 checks on tbl_Financials structure. (4) Management Report Tie-Out — 8 checks confirming Table = Dynamic NP outputs for Group and each entity, consolidation = sum of entities, source SUMIFS tie, period count. (5) Actual Period Integrity — 6 checks on non-zero actuals months, last actual month, Aug–Dec 2026 zeroed. (6) Source Data Integrity — 8 checks on account-level and entity-level SUMIFS vs report totals. (7) Dynamic Report Cross-Checks — 6 checks on entity NP ties, variance integrity. (8) Driver Forecast Integrity — 5 checks on historical tie, P&L arithmetic, consolidation, output tie, toggle response. Each check has: Description | Expected (blue hardcoded) | Actual (formula only) | Status =IF(Expected=Actual,"PASS","FAIL"). Section summaries and overall "X of 53 checks passed" at top. PASS = green fill, FAIL = red fill.

Prompt S2 — Management report – Table and Dynamic

Build two P&L management report sheets from tbl_Financials producing identical output. 'Mgmt Report — Table': SUMIFS approach with 36 monthly columns (Jan 2024 – Dec 2026), 4 entity blocks (Group Consolidated using EntityKey="*", Americas, APAC, EMEA), each with Actuals (ScenarioKey=1), Budget (ScenarioKey=2), Variance (=Actuals - Budget) sections, 12 P&L rows (Product Revenue, Services Revenue, Total Revenue, COGS, Gross Profit, S&W, Marketing, Software & Tools, T&E, Other OpEx, Total OpEx, Net Profit). Subtotals use SUM not SUMIFS. 'Mgmt Report — Dynamic': Same output using PLSTRUCTURE(), PLDATA(), PLDATABY() LAMBDA's from 'Dynamic Extracts'. One spilling formula per data block. Variance = Actuals# - Budget#. Both reports must match exactly.

Prompt S3 — Forecast accuracy

Build a 'Forecast Accuracy Report' for H1 CY2026 — January to June 2026 ONLY (6 months, DateKeys 20260131 to 20260630). Do NOT include July. 4 entity blocks (Consolidated, Americas, APAC, EMEA), each showing monthly Actuals, Budget, and Forecast rows, H1 Total, \$ Variance (Actuals minus Budget, and Actuals minus Forecast), and Accuracy % using =IFERROR(1-ABS(Variance/Budget),0). Use PLDATA/PLDATABY LAMBDA's filtered to the H1 date range. Accuracy >= 95% green highlight, < 80% red highlight.

Prompt S4 — Scenario modelling and executive summary

Build 5 interconnected sheets — all forecast cells must reference the Scenario Manager so one assumption change updates the entire model. (1) 'Scenario Manager' — 11 growth drivers across Base/Upside/Downside scenarios with July 2026 seed values from tbl_Financials and an Active column using INDEX/MATCH linked to 'Scenario Dashboard'!T22 toggle. (2) 'Driver Analysis' — Volume × Rate decomposition over 18 months (Feb 2026 – Jul 2027: 6 actuals + 12 forecast). Product Revenue = Volume × Unit Price, Services = Hours × Rate. Forecast compounds monthly: prior month × (1 + annual rate/12). COGS = Revenue × COGS%. All expenses negative. 3 entity blocks + Consolidated. (3) 'Forecast Output' — Monthly P&L from Driver Analysis consolidated with scenario comparison: Budget vs Original Forecast vs New Forecast. (4) 'Scenario Dashboard' — Dropdown toggle in T22 (Base/Upside/Downside) driving the Scenario Manager. Revenue and Net Income comparison charts with 4 series. (5) 'Executive Summary' — KPI tiles (FY26 Revenue, Gross Margin %, Net Profit, Budget/Forecast Accuracy %), FY P&L (FY25 vs FY26 vs Budget), entity breakdown, 4 charts.